

Plot analysis (GPT-4o)

The plot shows the yield stress versus temperature for three different Nb-Mo-Ta alloy compositions: Nb33Mo33Ta34, Nb20Mo20Ta60, and Nb20Mo60Ta20. Here's an analysis of the data:

Yield Stress Variation with Temperature

1. **General Trend**:

- For all three alloys, the yield stress decreases as the temperature increases. This is a typical behavior for metals, as higher temperatures generally lead to increased atomic vibrations, reducing the resistance to dislocation motion.

2. **Comparative Analysis**:

- **Nb20Mo20Ta60 (Green Squares)**:** This alloy exhibits the highest yield stress across the entire temperature range. It starts at around 950 MPa at low temperatures and decreases to about 200 MPa at 1000 K.
- **Nb33Mo33Ta34 (Blue Circles)**:** This alloy has a moderate yield stress, starting at approximately 800 MPa and decreasing to about 150 MPa at 1000 K.
- **Nb20Mo60Ta20 (Red Triangles)**:** This alloy shows the lowest yield stress, beginning at around 600 MPa and dropping to about 100 MPa at 1000 K.

Effects of Alloy Composition

- **High Ta Content (Nb20Mo20Ta60)**:** The high concentration of Ta (60%) significantly enhances the yield stress, likely due to Ta's ability to provide solid solution strengthening. Ta atoms create lattice distortions that impede dislocation motion, increasing the yield stress.
- **Balanced Composition (Nb33Mo33Ta34)**:** With nearly equal concentrations of Nb, Mo, and Ta, this alloy shows moderate strengthening. The balanced composition may lead to a more uniform distribution of solute atoms, providing moderate lattice distortion and strengthening.
- **High Mo Content (Nb20Mo60Ta20)**:** The high Mo concentration results in the lowest yield stress. While Mo can contribute to solid solution strengthening, its effectiveness may be less than that of Ta in this alloy system.

Mechanisms of Solute Strengthening

- **Solid Solution Strengthening**:** The primary mechanism at play is likely solid solution strengthening, where the different atomic sizes of Nb, Mo, and Ta create lattice distortions. These distortions hinder dislocation movement, increasing the yield stress.
- **Temperature Effects**:** As temperature increases, the effectiveness of solid solution strengthening decreases due to enhanced atomic mobility, which allows dislocations to bypass obstacles more easily.

Insights

- The alloy with the highest Ta content (Nb20Mo20Ta60) shows the greatest resistance to deformation, indicating Ta's significant role in strengthening.
- The decrease in yield stress with temperature suggests that these alloys may be more suitable for applications at lower temperatures where higher strength is required.
- Understanding the balance of alloying elements can help in designing alloys with desired mechanical properties for specific applications.

Figure S7: Results of the analysis conducted by the multi-modal agent powered by GPT-4o for the plot shown in Figure 7(d).